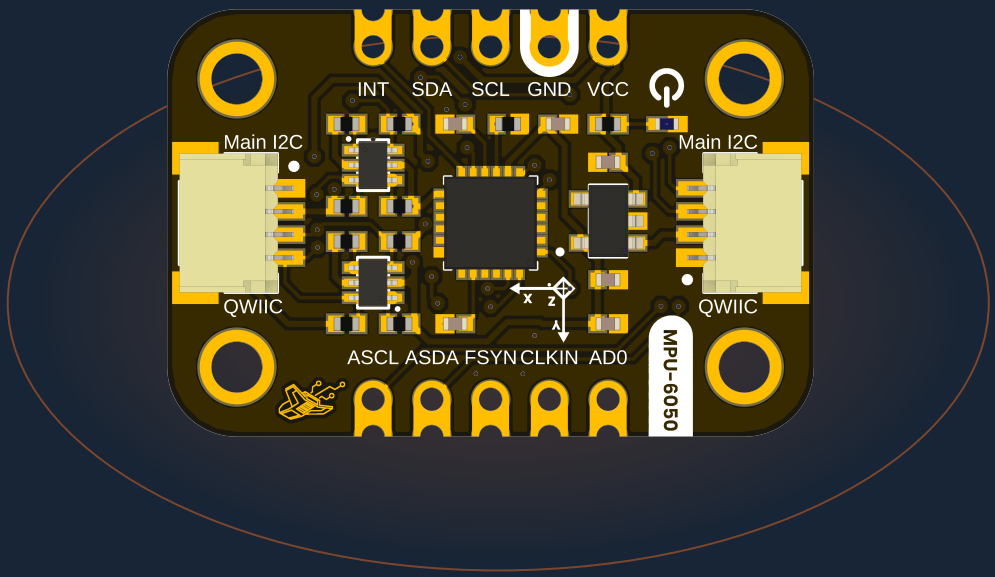
PRODUCT REFERENCE MANUAL

DEVLAB I2C INERTIAL SENSOR MODULE

DevLab I2C MPU-6050 6-Axis IMU Module

MPN: UE0129

Version 0.2.0



MPU-6050 module with three-axis acceleration and angular-rate sensing, main and auxiliary I2C connections, and onboard 3.3 V regulation.

I2C | 6-AXIS IMU | ACCELEROMETER | GYROSCOPE

DOCUMENT MAP

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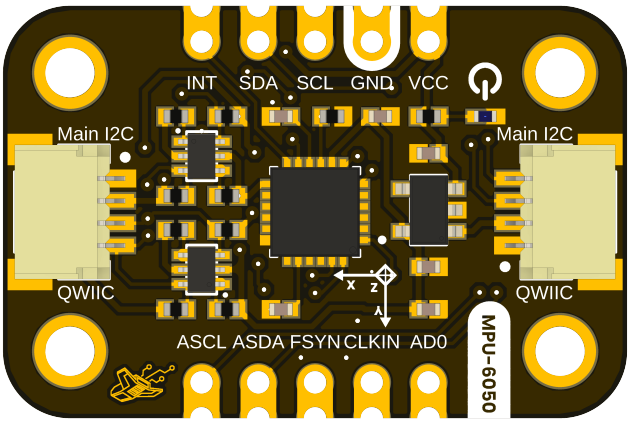
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Description

The UNIT DevLab I2C MPU-6050 6-Axis IMU Module combines a three-axis accelerometer and a three-axis gyroscope. The board routes the MPU-6050 main and auxiliary I2C buses, interrupt and synchronization signals, and includes 3.3 V regulation and I2C level shifting.



Top view of the UE0129 module.

Applications

- Evaluation of acceleration and angular-rate sensing
- Motion-sensing prototypes
- I2C sensor integration and educational experiments
- Experiments using interrupt, synchronization, and auxiliary-sensor signals

Hardware Features

- MPU-6050 three-axis accelerometer and three-axis gyroscope
- Six 16-bit ADCs, one for each accelerometer and gyroscope axis
- Selectable accelerometer ranges of ±2 g, ±4 g, ±8 g, and ±16 g
- Selectable gyroscope ranges of ±250, ±500, ±1000, and ±2000 dps
- 1024-byte sensor FIFO and digital-output temperature sensor
- Main I2C bus on two board connectors and an edge connection
- Auxiliary I2C bus on two board connectors and an edge connection
- `INT`, `AD0`, `FSYNC`, and `CLKIN` signals
- AP2112K-3.3 fixed 3.3 V regulator
- Two BSS138AKDW dual N-channel MOSFET arrays for I2C level shifting
- Red power indicator
- Four mounting holes and ten edge connections shown by the board artwork

1 The Module

1.1 Board Identification

Item	Value
Product	DevLab: I2C MPU-6050 6-Axis IMU Module
Manufacturer part number	UE0129
Hardware revision	V1.0
Inertial sensor	MPU-6050
Host interface	I2C
Product-reference revision	0.2.0

1.2 Package Contents

The BOM includes one four-pin, 1 mm-pitch harness. Package contents and cable compatibility have not been formally validated; confirm them with the distributor before purchase or integration.

2 Electrical Characteristics

2.1 Confirmed Circuit Values

Symbol / parameter	Description	Value	Scope
VDD	Operating supply	2.375 to 3.46 V	MPU-6050 sensor pins
VLOGIC	Logic reference	1.71 V to VDD	MPU-6050 sensor pin
Accelerometer range	Selectable full scale	±2, ±4, ±8, or ±16 g	MPU-6050 sensor
Gyroscope range	Selectable full scale	±250, ±500, ±1000, or ±2000 dps	MPU-6050 sensor
ADC word length	Per accelerometer / gyroscope axis	16 bit	MPU-6050 sensor
I2C operating frequency	Fast-mode maximum	400 kHz	MPU-6050 sensor interface
Six-axis current	Gyroscope + accelerometer, DMP disabled	3.8 mA typical	MPU-6050 sensor only
Six-axis + DMP current	Gyroscope + accelerometer + DMP	3.9 mA typical	MPU-6050 sensor only
3V3	Regulated output from U2	3.3 V nominal	AP2112K-3.3 regulator output
C1, C2	Regulator input/output capacitors	1 µF each	Individual BOM components
R5	Power-indicator series resistor	4.7 kΩ	Individual BOM component
R1–R4, R6–R10	Pull-up / bias resistors	10 kΩ each	Individual BOM components

The allowed voltage at the module VCC connections, module current consumption, host-side digital thresholds, and complete-module I2C clock limit are **pending validation**. The sensor, regulator, connector, MOSFET, resistor, and capacitor component ratings are not ratings for the complete module.

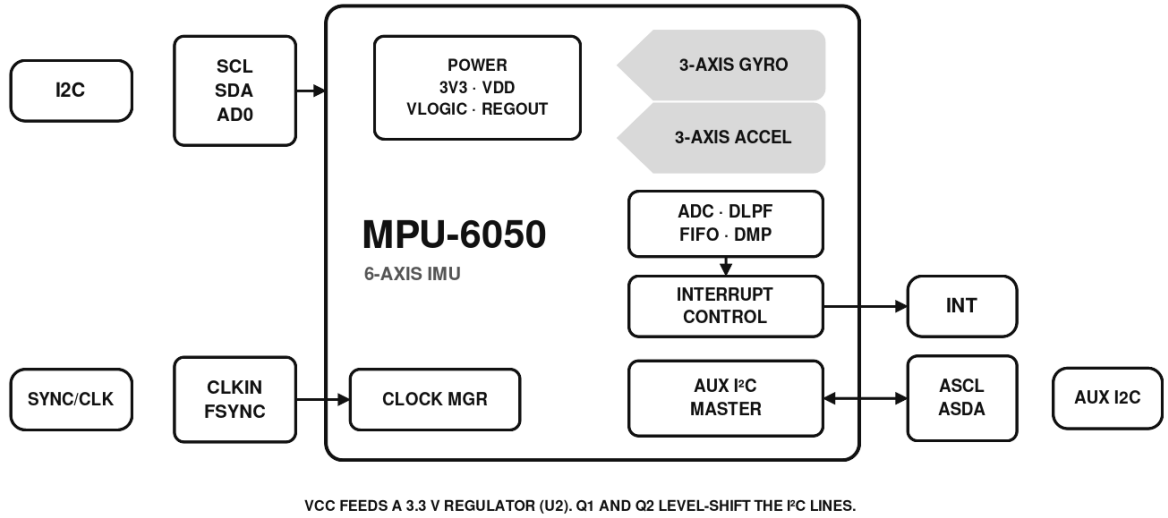
2.2 Electrical Precautions

- Establish a common GND before applying power or interface signals.
- Verify the module-level VCC range before applying power.
- Keep host signals within validated logic levels; those limits are not yet documented for the complete module.
- Treat INT as an output from the sensor and CLKIN, FSYNC, and AD0 as sensor inputs according to the released schematic.

3 Functional Overview

3.1 Block Diagram

The block diagram shows the main functional blocks of the MPU-6050 DevLab module and the signals available.



MPU-6050 block diagram.

3.2 Sensor and Main I2C Path

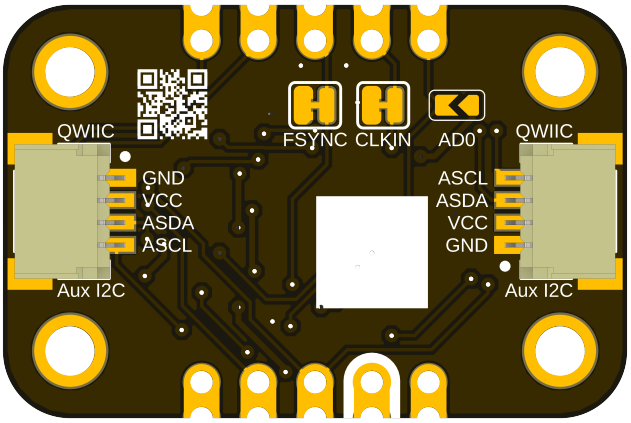
The MPU-6050 provides three-axis acceleration and three-axis angular-rate sensing with 16-bit conversion per axis and selectable full-scale ranges. Its 1024-byte FIFO can buffer sensor data. Q1 translates the main SCL and SDA lines between the module VCC domain and the sensor's regulated 3.3 V domain. J1, J3, and J2 provide access to the main bus.

3.3 Auxiliary and Control Signals

Q2 translates the MPU-6050 auxiliary AUX_SCL and AUX_SDA lines to the board ASCL and ASDA connections. The board also routes INT, AD0, FSYNC, and CLKIN. The schematic shows R6 connected from AD0 to ground, and the module's default I2C address is 0x68. The physical defaults of the FSYNC and CLKIN solder options shown in the artwork remain pending validation.

3.4 Power Section

The schematic identifies U2 as an AP2112K-3.3 regulator supplied from VCC. Its 3.3 V output powers both MPU-6050 supply pins and the sensor-side I2C pull-ups. The complete module input range is pending validation and must not be derived from the regulator's component rating.



Bottom view of the UE0129 module, artwork V1.1.

3.5 Communication Interfaces

The MPU-6050 supports a single digital communication interface, I²C, providing connectivity with microcontrollers and embedded systems.

I²C Interface

- Standard Mode and Fast Mode
- Clock frequency up to 400 kHz
- Two selectable 7-bit I²C addresses: `0x68` and `0x69`
- I²C address selection through the `AD0` pin

The device supports two selectable 7-bit I²C addresses through the `AD0` pin: when `AD0` is low, the address is `0x68`; when `AD0` is high, the address is `0x69`. On the DevLab module, `AD0` is routed to the J5 connector and has a 10 kΩ resistor (R6) to ground. The module's default I²C address is `0x68`.

3.6 Library Features

The DevLab_MPU6050 library for Arduino-compatible boards communicates with the sensor through direct I²C register access. It is tested on ESP32 and compatible with other Arduino-compatible boards. The following table summarizes the sensor functions available through the library.

Function	Library call	Description
Initialization	`begin(sda, scl, clock)`	Starts I²C on custom pins (400 kHz by default), verifies `WHO_AM_I`, wakes the sensor, and applies the default configuration.
Data acquisition	`readData(data)`	Reads raw 16-bit accelerometer and gyroscope values for X, Y, and Z in a single 14-byte burst, plus the temperature in °C.
Accelerometer range	`setAccelScale(scale)`	Selects ±2 g, ±4 g, ±8 g, or ±16 g (values 0 to 3).
Gyroscope range	`setGyroScale(scale)`	Selects ±250, ±500, ±1000, or ±2000 °/s (values 0 to 3).

Function	Library call	Description
Digital low-pass filter	`setDLPF(setting)`	Selects the DLPF configuration in the `CONFIG` register (values 0 to 6).
Sample rate	`sampleRateDiv(div)`	Sets the sample rate divider (`SMPLRT_DIV`).
Clock source	`setClkSource(source)`	Selects the clock source in `PWR_MGMT_1` (values 0 to 7).
Device identification	`WhoAmI()`	Returns the `WHO_AM_I` register value.
Interrupt pin	`setIntPinCfg(cfg)`	Configures `INT` polarity, output type, latch, clear-on-read, `FSYNC`, and I^2C bypass.
Interrupt sources	`enableInterrupt()`, `disableInterrupt()`	Enables or disables data ready, I^2C master, FIFO overflow, and motion interrupts.
Interrupt status	`interruptStatus()`	Reads `INT_STATUS` to identify which source fired.

The I^2C address is passed to the library constructor and is not detected automatically. Use `MPU6050_ADDRESS_AD0_LOW` (`0x68`) or `MPU6050_ADDRESS_AD0_HIGH` (`0x69`) according to the `AD0` level. The `WHO_AM_I` check in `begin()` expects `0x68` regardless of the selected address.

Default configuration applied by `begin()`

Parameter	Default value
Clock source	PLL with X-axis gyroscope reference
Sample rate divider	7
Digital low-pass filter	Configuration 6
Gyroscope range	±500 °/s
Accelerometer range	±8 g

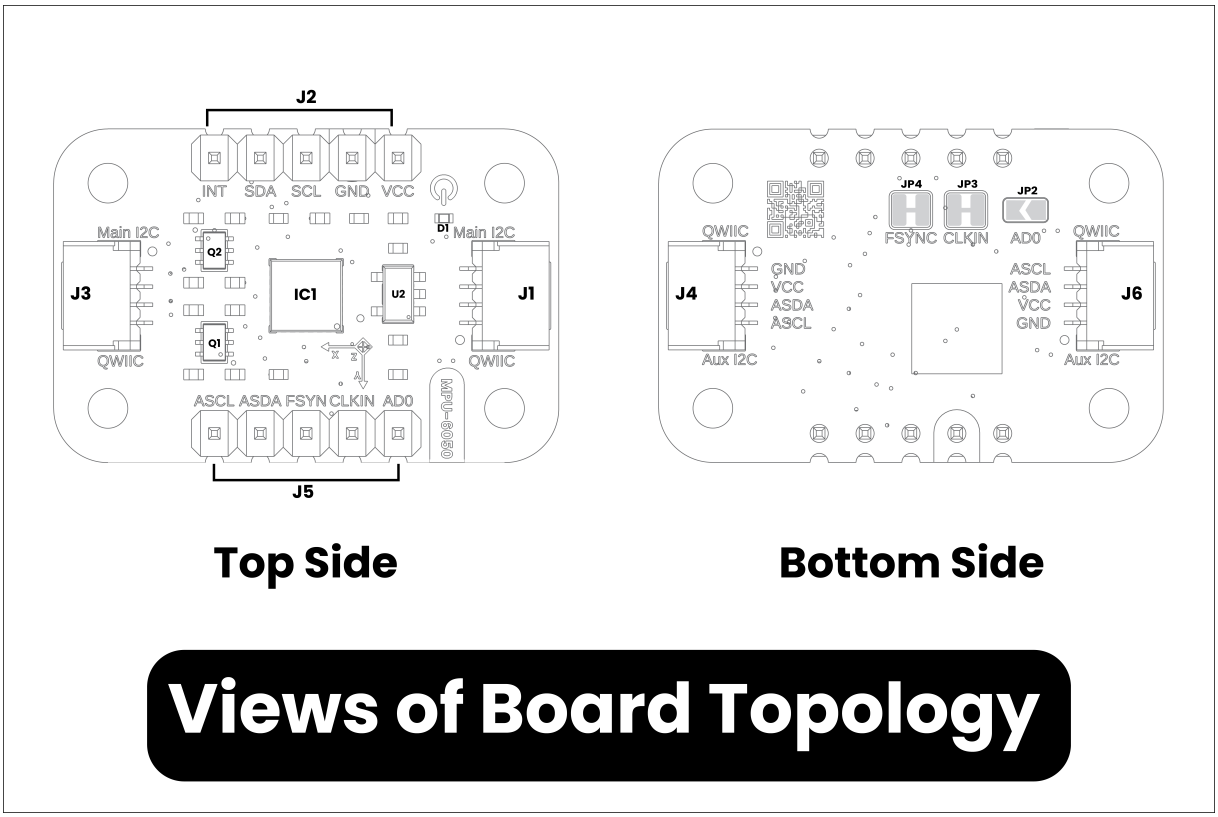
Use the configuration functions above to change these values after calling `begin()`.

Example sketches

- `BasicRead` and `MPU6050_BasicRead`: read acceleration, angular rate, and temperature and print them over the serial port.
- `I2CIdentify`: initializes the sensor and reads the `WHO_AM_I` register.
- `MPU6050_ConfigSweep`: sweeps every combination of accelerometer range, gyroscope range, and DLPF setting and prints one reading per combination as CSV, useful for characterizing each setting.
- `basic_Interrupt` and `MPU6050_Interrupt`: enable the data-ready interrupt on the `INT` pin and count the interrupts it generates.

3.7 Board Topology

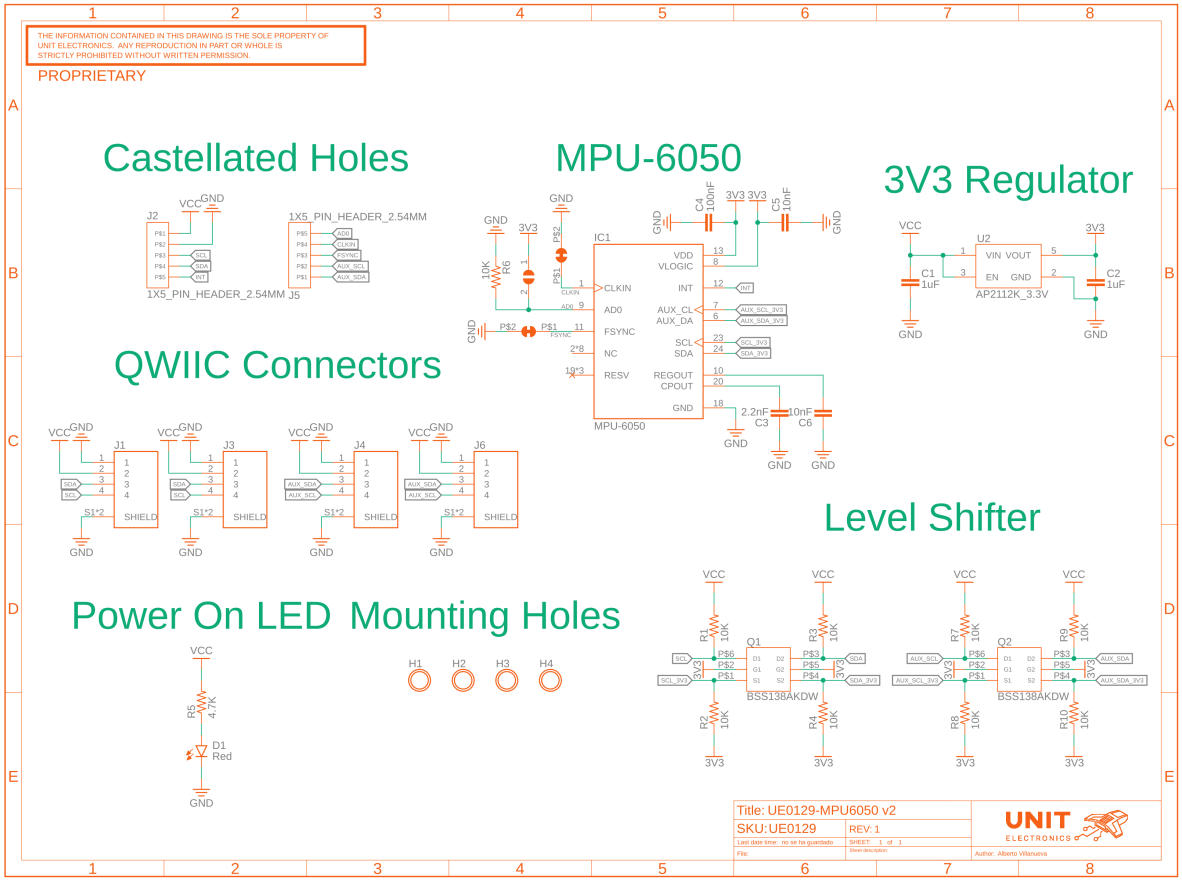
The released topology drawing locates the sensor, regulator, level shifters, connectors, and solder options on the top and bottom sides of the board.



Top and bottom board topology.

3.8 Electrical Schematic

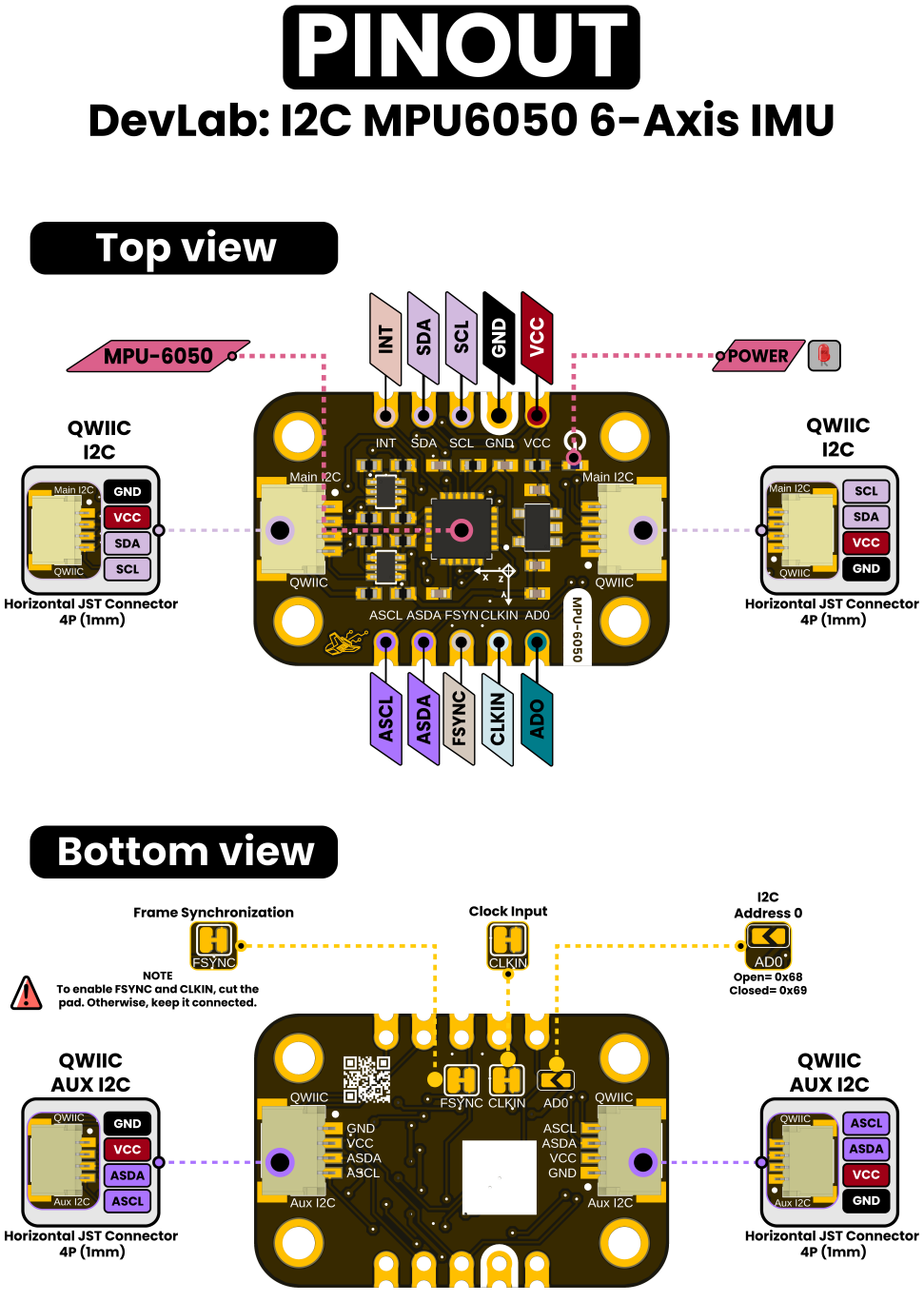
The image below is rendered from the released V1.0 schematic. Use the original schematic PDF to inspect small component and net labels at full resolution.



V1.0 electrical schematic.

4 Connectors and Pinout

The released pinout drawing shows the top and bottom board views and the connector signal labels. Confirm connector orientation on the physical board before assigning numeric pin positions or connecting a cable.



UE0129 pinout, top and bottom views.

4.1 Main Interface Edge (J2)

Board label	Direction / type	Function
VCC	Power input	Module input rail; allowed range pending validation

Board label	Direction / type	Function
GND	Power	Common ground
SCL	I2C clock	Main host-side I2C clock
SDA	I2C data	Main host-side I2C data
INT	Output	MPU-6050 interrupt signal

4.2 Auxiliary and Control Edge (J5)

Board label	Direction / type	Function
AD0	Input	MPU-6050 I2C address-selection input
CLKIN	Input	External clock input to the MPU-6050
FSYNC	Input	Frame-synchronization input
ASCL	I2C clock	Host-side auxiliary I2C clock
ASDA	I2C data	Host-side auxiliary I2C data

4.3 Four-Position Connectors

Reference	Signals	Bus
J1, J3	VCC, GND, SDA, SCL	Main I2C
J4, J6	VCC, GND, ASDA, ASCL	Auxiliary I2C

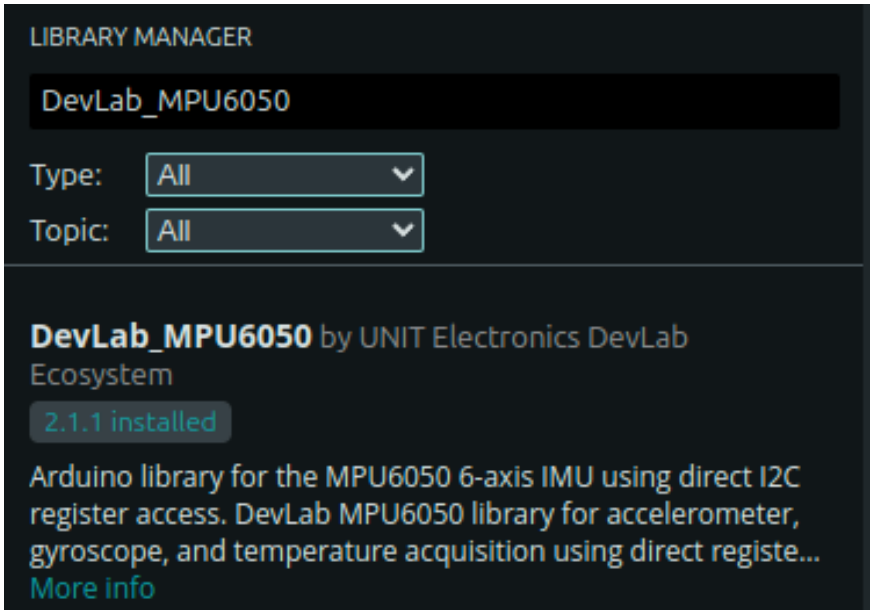
The BOM identifies all four as JST SM04B-SRSS-TB(LF)(SN) connectors. The artwork labels them QWIIC, but connector orientation, numeric pin numbering, and cable compatibility must be verified before assembly or connection.

4.4 Solder Options

The bottom artwork identifies options for FSYNC, CLKIN, and AD0. Their default copper states and intended operating configurations are pending validation.

5 Board Operation

The DevLab MPU-6050 module can be integrated into Arduino-compatible development boards through the I²C communication interface. Before running any application, install the DevLab_MPU6050 library from the Arduino Library Manager or download it from the official GitHub repository.



DevLab_MPU6050 library in the Arduino Library Manager.

Hardware Interface

I²C Connection

Connect the module to the host board using the main I²C interface, available through the J2 header or the J1/J3 QWIIC connectors:

Module signal	Description	Host connection
VCC	Power supply	Host supply (approved range pending validation)
GND	Ground	Host ground
SDA	I²C data	Host SDA pin
SCL	I²C clock	Host SCL pin

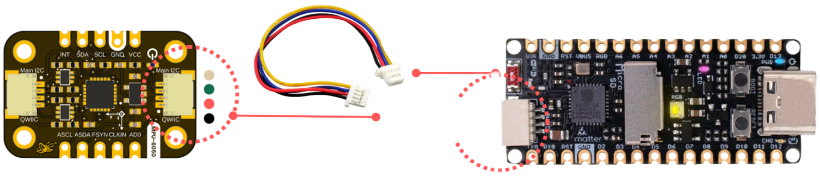
The INT pin is optional and is only needed for interrupt-driven applications. Host GPIO numbers belong to the controller, not the module. The example below uses GPIO 6 for SDA and GPIO 7 for SCL on an ESP32 board(you can modify for your specific board):

```
#define SDA_PIN 6
#define SCL_PIN 7
```

Default I²C Address

Device	Address
MPU-6050	`0x68`

The module's default 7-bit I²C address is `0x68`. The sensor datasheet assigns `0x68` when `AD0` is low and `0x69` when it is high.



MPU-6050 module connected to the host board through I²C.

Example Operation

After `begin()`, the library applies the following configuration:

Parameter	Resulting value
Clock source	PLL with X-axis gyroscope reference
Sample rate divider	7
Digital low-pass filter	Configuration 6
Gyroscope full scale	±500 °/s
Accelerometer full scale	±8 g
I²C clock	400 kHz

Basic Operation

The sensor reads three groups of values:

Value	Description
Accelerometer	Raw 16-bit acceleration for each axis (`accX`, `accY`, `accZ`)
Gyroscope	Raw 16-bit angular rate for each axis (`gyrX`, `gyrY`, `gyrZ`)
Temperature	Internal die temperature, in °C (`temperatureC`)

The accelerometer and gyroscope values are raw counts; their scale depends on the selected full-scale range.

1. Connect the module to the microcontroller using the I²C interface.
2. Open the Arduino IDE.
3. Install the DevLab_MPU6050 library.
4. Open the `BasicRead` example sketch provided with the library.
5. Select the correct board and serial port.
6. Compile and upload the example.
7. Open the Serial Monitor to display the accelerometer, gyroscope, and temperature measurements.

During `begin()`, the library verifies communication by reading the `WHO_AM_I` register (expected value `0x68`), wakes the sensor, and applies the default configuration. If initialization fails, check the wiring, the supply, and the I²C address.

Developers can modify the example code to configure:

- Accelerometer measurement range
- Gyroscope measurement range
- Digital low-pass filter
- Sample rate
- Clock source
- Interrupt generation

Example Code

Use the `BasicRead` example included with the library, or copy the following sketch into a new sketch:

```
#include <DevLab_MPU6050.h>

#define SDA_PIN 6
#define SCL_PIN 7
#define MPU6050_ADDR 0x68

DevLab_MPU6050 imu(Wire, MPU6050_ADDR);

SensorData data;

void setup() {
  Serial.begin(115200);
  delay(1000);

  Serial.println("Inicializando MPU6050...");

  if (!imu.begin(SDA_PIN, SCL_PIN, 400000)) {
    Serial.println("ERROR: MPU6050 no encontrado.");
    while (1) {
      delay(1000);
    }
  }

  Serial.println("MPU6050 inicializado correctamente.");
}

void loop() {
  if (imu.readData(data)) {
    Serial.print("aX = ");
    Serial.print(data.accX / 4096.0f);
    Serial.print(" | aY = ");
    Serial.print(data.accY / 4096.0f);
    Serial.print(" | aZ = ");
    Serial.print(data.accZ / 4096.0f);

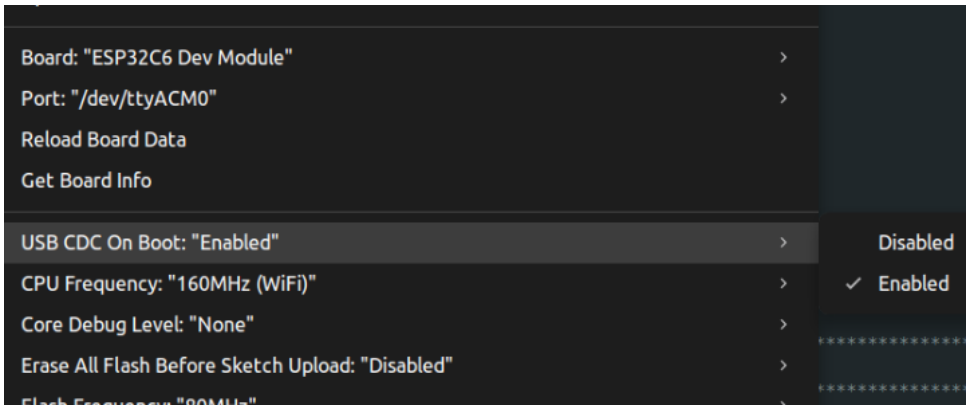
    Serial.print(" | temp = ");
    Serial.print(data.temperatureC, 2);

    Serial.print(" | gX = ");
    Serial.print(data.gyrX);
    Serial.print(" | gY = ");
    Serial.print(data.gyrY);
    Serial.print(" | gZ = ");
    Serial.println(data.gyrZ);
  } else {
    Serial.println("ERROR: fallo en la lectura del MPU6050.");
  }

  delay(1000);
}
```

Important

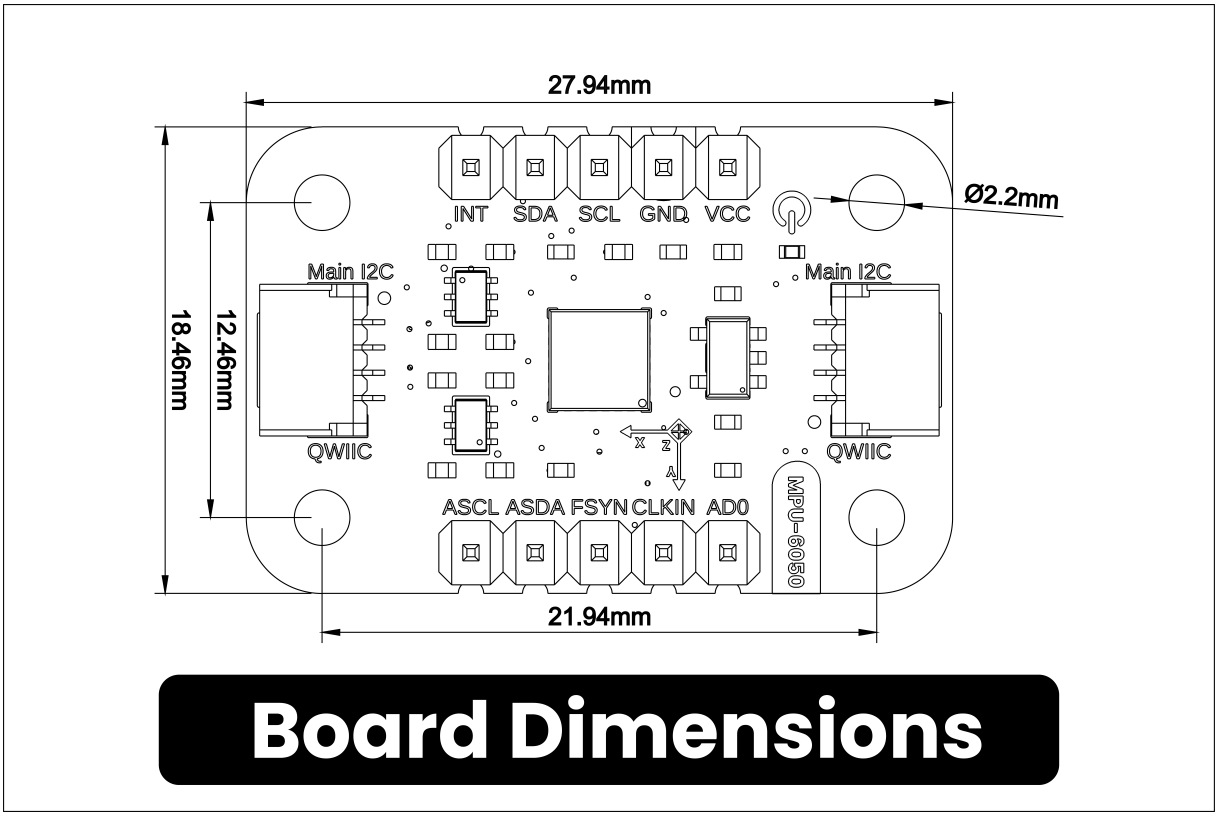
Make sure **USB CDC On Boot** is set to **Enabled**. This allows the ESP32-C6 to expose the USB Serial interface after boot, so the `Serial` output can be viewed in the Arduino Serial Monitor. If this option is disabled, the sketch will still run, but you may not see any serial messages.



USB CDC On Boot option enabled in the Arduino IDE.

Module-level I^2C clock validation and the approved VCC range remain pending testing.

6 Mechanical Information



Released board dimensions drawing.

The released drawing labels a 27.94 × 18.46 mm board outline, four Ø2.2 mm mounting holes, and 21.94 × 12.46 mm hole-center spacing. The drawing also shows ten castellated edge connections. These are drawing values; mechanical tolerances and fit remain **pending validation**. Use the labeled dimensions rather than scaling the image.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	https://uelectronics.com/
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

Product specifications and documentation may be revised. Use the repository and product page listed in the next chapter for the latest released material.

8 Reference Documentation

Reference	Location
Repository	UNIT-Electronics-MX/unit_devlab_i2c_mpu6050_6_axis_imu
Hardware notes	Hardware README
Sensor datasheet	MPU-6000 and MPU-6050 Product Specification, Revision 3.4
Module schematic	V1.0 schematic
Schematic image	V1.0 schematic PNG
Pinout	English PDF · Spanish PDF
Pinout images	English PNG · Spanish PNG
Board topology	Top and bottom topology drawing
Board dimensions	Dimensioned drawing
Manufacturing BOM	V1.0 BOM
Experimental firmware	DevLab_MPU6050 basic readout

Sensor values in this reference are taken from the MPU-6050 portions of the Revision 3.4 product specification. Module-level limits require separate board validation.

9 Appendix

9.1 Source Hierarchy

This product reference uses the MPU-6000 and MPU-6050 Product Specification, Revision 3.4 for sensor characteristics, the released schematic for circuit connectivity, the manufacturing BOM for fitted component identities and values, and the released pinout, topology, board views, and dimensioned drawing for visible labels and physical features. A future approved validation report takes precedence if it conflicts with this edition.

9.2 Items Pending Validation

- Module VCC operating and absolute-maximum ratings
- Module current consumption and host-side logic levels
- Module-level I2C clock testing, board-default address, and configuration sequences
- Mechanical tolerances and physical fit against the dimensioned drawing
- Solder-option defaults and connector orientation
- Cable and ecosystem compatibility
- Validated firmware examples and measurement results

9.3 Document Control

Field	Value
Product	DevLab: I2C MPU-6050 6-Axis IMU Module
Manufacturer part number	UE0129
Hardware revision	V1.0
Product-reference revision	0.2.0
Publication date	2026-10-21
Firmware status	Experimental; not validated by this edition